Variables

13 June 2026 10:25

What is variable ?

In real life we store things in containers

EG:

1. Water bottle stores water
2. School bag stores books
3. Wallet stores money

Similarly in java a variable is also like container

Definition

> **Variable** = a named box/container used to store values in java



→ `sandook`



→ `bag`



→ `bottle`

```
int num=20;
```

- `int` means the type of value is an integer value
- **num :** is the name of variable in java. This name is also called an **identifier**.

Why we need variables ?

Without variables

```
System.out.println(20);
System.out.println(20);
System.out.println(20);
System.out.println(20);
```

If num changes from 20 to 21 we must change it everywhere.

With variable

```
int num=20;
System.out.println(num);
System.out.println(num);
System.out.println(num);
```

Now if num changes from 20 to 21, we only change it at one place

Basic syntax of variable

```
dataType variableName = value
```

EG:

```
int marks=85;
```

int	marks	=	85	;
data type	variable name	operator	value	statement end

Q1 **Write java program to print age 20:**

Variable declaration and initialization

```
int age=20;
```

Q2 **Write java program to take one age variable and print it, then change its value and print it**

> **Note:** variable name (identifiers) have some rules

Rules:

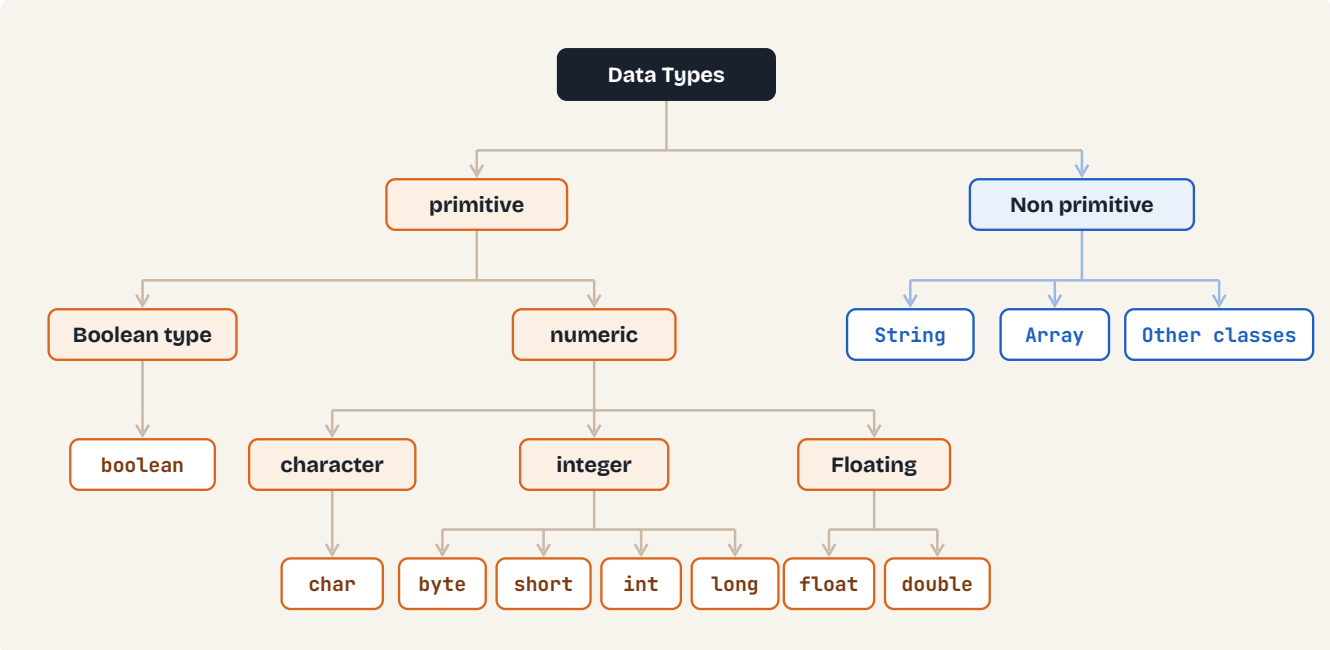
1. Lower case, Upper case, digit, `$` , `_`
2. **Start with :** letter, `_` , `$` . It cannot start with digit
3. **Case sensitive:** `age` , `Age` , `AGE` `AgE`
4. **Reserved keywords :** `int` , `for` , `if` cannot be used

Examples:

1. `Age` :
2. `_marks`
3. `$value`
4. `1cityName` :
5. `cityName` :
6. `#myname` :

BREAK : 11:20

Data Types



Before understanding data types, first understand data

Name:	`rahul`
Age:	`20`
Marks :	`50`
Price:	`77.80`
Grade :	`A`
Payment done:	`true`

What is Data type ?

> **Data type** = type/category of data

```
int age=20
```

- **20** = integer type ki value
- **age** = var name

Variable is the box
Data type tells what kind of item we can put inside the box

Java has 8 primitive data types:

Data type	Used for	Example
`byte`	Small whole number	`100`
`short`	Whole	`30000`
`int`	Common whole num	`23`
`long`	Very large whole num	`9989788686677`
`float`	Decimal num	`10.5f`
`double`	Common decimal num	`99.99`
`char`	Single char	`'A'`
`boolean`	True/false	`true`

Data Types	Default Size	Default Value	Comments
`byte`	1 Byte	`0`	Best suited for Stream I/O operation
`short`	2 Bytes	`0`	Rarely used nowadays
`int`	4 Bytes	`0`	It is the default data type of java to hold integer values
`long`	8 Bytes	`0L`	It is recommended for the files with size in GB's
`float`	4 Bytes	`0.0f`	For 5 point precision
`double`	8 Bytes	`0.0D`	For 15 point precision
`char`	2 Bytes	`'\u0000'`	It is of 2 Bytes because it supports I18N
`boolean`	1 bit	`false`	It stores the literal true or false only

